

NexGen NeuroVision Project

Report

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Executive Summary

This document evaluates the operational performance differences between **spatial** and **sequential neural networks** applied to raw medical imagery. Across three distinct architectures benchmarked against a comprehensive dataset of brain magnetic resonance imaging (MRI) scans, the empirical data establishes that **spatial convolutional networks fundamentally outperform sequential models for pixel-based classification tasks**.

Dataset & Preprocessing

We sourced **7,200 perfectly balanced MRI scans** across four distinct categories: **glioma**, **meningioma**, **pituitary**, and **no tumor** cases. Data was partitioned into training, validation, and testing sets using a **66:12:22 ratio** to ensure absolute isolation and prevent data leakage.

To combat overfitting, the training pipeline deployed dynamic image augmentation including random horizontal flips, rotational shifts, and color adjustments. All images were resized to **224×224 pixels** and normalized against standard ImageNet distribution metrics. Validation and test sets remained completely unaltered to preserve ground truth integrity.

Architectural Methodology

We engineered three separate architectures to test spatial versus sequential feature extraction paradigms.

- **CNN ResNet50:** Deployed a **ResNet50** architecture leveraging transfer learning with locked base weights and a custom multi-layer classification head with heavy dropout to map spatial features directly to the four tumor classes.
- **RNN Sequential:** Constructed a sequential model reading images row-by-row, flattening images into sequences of **224 time steps** to process rows sequentially for temporal dependency extraction.
- **LSTM Sequential:** Enhanced the standard recurrent network by replacing basic cells with memory gating mechanisms to mitigate gradient loss across the long **224-step sequence length**.

Performance Metrics

Empirical telemetry demonstrates a distinct performance advantage for spatial models over sequential architectures across all primary key performance indicators:

Model	Accuracy	Precision	Recall	F1 Score	Parameters	Training Time (s)
CNN ResNet50	0.82875	0.831101	0.82875	0.822492	525,572	726.46
RNN Sequential	0.56875	0.554586	0.56875	0.516123	144,196	484.25
LSTM Sequential	0.704375	0.725500	0.704375	0.683947	551,236	516.12

Visual Analytics

Area Under the Curve (AUC) Analysis

The macro-average Receiver Operating Characteristic (ROC) curves confirm overall classification reliability. **CNN ResNet50** leads with an extraordinary AUC of **0.9626**, followed by **LSTM Sequential** at **0.8995**, and **RNN Sequential** trailing at **0.8214**.

Error Topography & Confusion Matrices

Confusion matrix evaluation highlights clear operational failure modes in sequential processing. **CNN ResNet50** accurately identifies the vast majority of no tumor and pituitary cases, showing minimal confusion between glioma and meningioma. Conversely, **LSTM Sequential** struggles to isolate meningioma, while **RNN Sequential** fails significantly across spatial boundaries.

Learning Progression & Convergence

Training loss and validation accuracy curves show that **CNN ResNet50** achieves stable convergence with validation accuracy peaking near **90%**. **LSTM Sequential** plateaus early, and **RNN Sequential** exhibits high volatility, failing to generalize beyond **60% accuracy**.

Final Conclusions

The empirical results conclusively prove that **CNN ResNet50 is the superior deployment architecture** for image-based diagnostics. Flattening images into 1D sequences destroys two-dimensional spatial context and edge boundaries vital for identifying tumor geometry.

Recommendation: Standardize all production image classification pipelines on spatial convolutional networks.